

(a) Relative tactic occurrence (%)

		APT attacker model				baseline attacker
		c_1	c_2	c_3	c_4	
SCAN_HOST	Reconnaissance	5	1	1	2	2
ENUM_HOST	Lateral movement	9	8	5	14	25
SCAN_PORT	Discovery	8	12	17	8	25
	Initial access	3	6	1	<1	23
EXPLOIT_VULN	Execution	8	14	9	17	
	Privilege escalation	3	3	7	3	
BRUTE_FORCE	Credential access	10	3	1	10	16
SCAN_NEIGHBOR	Command and control	17	14	11	19	9
	Resource development	0	0	1	3	—
	Persistence	8	13	10	4	
	Stealth	14	16	16	10	
<i>dwell-only</i>	Defense impairment	1	—	—	5	
	Collection	7	<1	4	4	
	Exfiltration	6	—	4	—	
	Impact	<1	8	12	—	

(b) Attack path variation (APV)